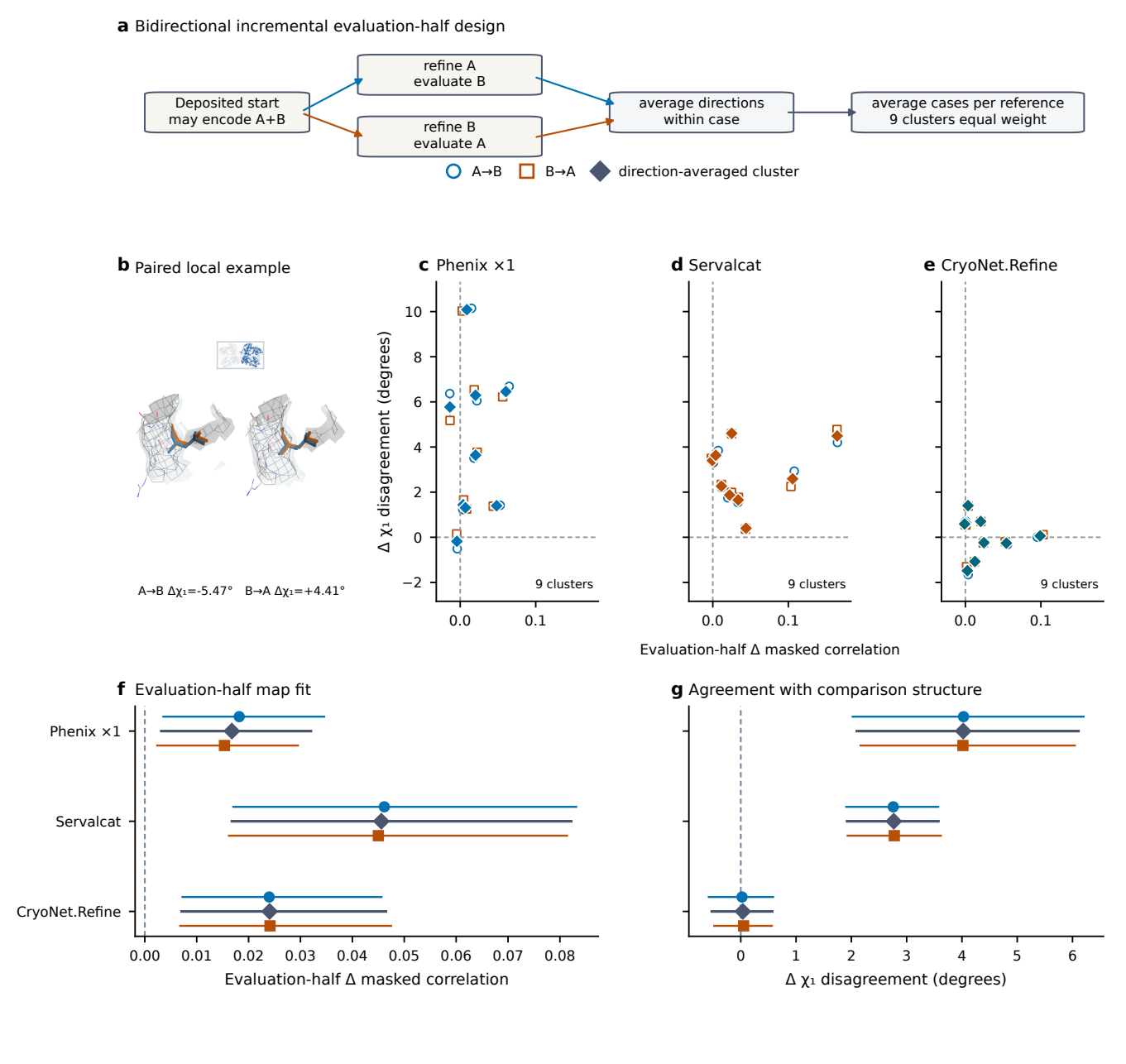

a Bidirectional incremental evaluation-half design
Deposited start may encode A+B
refine A evaluate B
refine B evaluate A
average directions within case
average cases per reference 9 clusters equal weight
A→B
B→A
direction-averaged cluster
b Paired local example
A→B Δχ1=-5.47°
B→A Δχ1=+4.41°
c Phenix ×1
d Servalcat
e CryoNet.Refine
Δ χ1 disagreement (degrees)
9 clusters
Evaluation-half Δ masked correlation
f Evaluation-half map fit
Phenix ×1
Servalcat
CryoNet.Refine
Evaluation-half Δ masked correlation
g Agreement with comparison structure
Δ χ1 disagreement (degrees)